

Recap

- NLS
- Metric space
- Examples
- Closed subspace of NLS.

> Relation

• Cauchy-Schwartz Ineq

$$\left| \sum_{i=1}^n x(i)y(i) \right| \leq \left(\sum_{i=1}^n |x(i)|^2 \right)^{1/2} \left(\sum_{i=1}^n |y(i)|^2 \right)^{1/2}$$

• Hölder's Ineq

$$\left| \sum_{i=1}^n x(i)y(i) \right| \leq \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} \left(\sum_{i=1}^n |y(i)|^q \right)^{1/q}$$

$p > 1$.

• Minkowski's Ineq

$$\left(\sum_{i=1}^n |x(i)+y(i)|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} + \left(\sum_{i=1}^n |y(i)|^p \right)^{1/p}$$

Examples.

$\mathbb{K}^n$

$p > 1$

Define $x \mapsto \|x\|_p$

$$\|x\|_p = \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p}, \quad x(i) \in \mathbb{K}$$

$$\cdot \quad \|x+y\|_p \leq \|x\|_p + \|y\|_p$$

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x(i)|$$

$p = \infty$

(Exercise $\|\cdot\|_p$ - $\|\cdot\|_\infty$ are norms) $p \in \mathbb{R}$.

$p \in [1, \infty)$

Q What happens when $p \notin [1, \infty]$?

$$\cdot \quad p = 0$$

X

$$\cdot \quad p \in \mathbb{R}^-$$

X

$$0 < p < 1$$

Then $\|\cdot\|_p$ defined above does not satisfy triangle ineq.

$$e_1 = (1, 0, \dots, 0)$$

$$e_2 = (0, 1, \dots, 0)$$

$$\|e_1\|_p = \left(\sum_{i=1}^n |e_1(i)|^p \right)^{1/p} = 1$$

$$\|e_2\|_p = 1$$

$$e_1 + e_2 = (1, 1, 0, \dots, 0)$$

$$\|e_1 + e_2\|_p = 2^{1/p}$$

Since $0 < p < 1$

$$2^{1/p} \geq 2$$

$$\|e_1 + e_2\|_p \geq \|e_1\|_p + \|e_2\|_p \xrightarrow{\text{triangle ineq.}}$$

doesn't hold.

- $\mathbb{K}^n \cong \mathcal{F}(S, \mathbb{K})$
 $S = \{1, 2 \dots n\}$

- $\mathcal{F}(\mathbb{N}, \mathbb{K}) = \text{set of seq. } S \neq \emptyset$
 ℓ -space.

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x(i)|^p \right)^{1/p}$$

$$\left(\sum_{i=1}^n |x(i) + y(i)|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} + \underbrace{\left(\sum_{i=1}^n |y(i)|^p \right)^{1/p}}$$

$\swarrow$

Observe

that if $\left(\sum_{i=1}^{\infty} |x(i)|^p \right)^{1/p} < \infty$ δ

$$\left(\sum_{i=1}^{\infty} |y(i)|^p \right)^{1/p} < \infty$$

the n a priori we cannot
say $\left(\sum_{i=1}^{\infty} |x(i) + y(i)|^p \right)^{1/p} < \infty$.

Consider,

$$S_N = \left(\sum_{i=1}^N |x(i) + y(i)|^p \right)^{1/p}.$$

- S_N is monotonically increasing.

$$\begin{aligned} \underline{S_N} &= \left(\sum_{i=1}^N |x(i) + y(i)|^p \right)^{1/p} \\ &\leq \left(\sum_{i=1}^N |x(i)|^p \right)^{1/p} + \left(\sum_{i=1}^N |y(i)|^p \right)^{1/p} \\ &\leq \left(\sum_{i=1}^{\infty} |x(i)|^p \right)^{1/p} + \left(\sum_{i=1}^{\infty} |y(i)|^p \right)^{1/p} \quad (\text{Minkowski's Ineq}). \\ &= \underline{K_1 + K_2} \end{aligned}$$

$$\cdot \left(\sum_{i=1}^{\infty} |x(i) + y(i)|^p \right)^{1/p} \leq \left(\sum_{i=1}^{\infty} |x(i)|^p \right)^{1/p} + \left(\sum_{i=1}^{\infty} |y(i)|^p \right)^{1/p} \\ < \infty \quad \text{---} *$$

$$l^p = \left\{ x \in F(\mathbb{N}, \mathbb{K}) \mid \sum |x(i)|^p < \infty \right\} \\ l^p \subseteq F(\mathbb{N}, \mathbb{K}) \xrightarrow{\quad} \text{lin space.}$$

$$\cdot x, y \in l^p \Rightarrow x + y \in l^p \leftarrow (\text{by } *) \\ \cdot x \in l^p, \lambda \in \mathbb{K} \Rightarrow \lambda x \in l^p.$$

l^p is a subspace of $F(\mathbb{N}, \mathbb{K})$.

For each $x \in l^p$ we define

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x(i)|^p \right)^{1/p} \quad 1 \leq p < \infty$$

Consider

$$l^{\infty} = \left\{ x \in F(\mathbb{N}, \mathbb{K}) \text{ such that } (x(n)) \text{ is bounded} \right\}$$

Clearly l^{∞} is a subspace of $F(\mathbb{N}, \mathbb{K})$.

$x \in l^{\infty}$ we defined

$$\|x\|_{\infty} = \sup_{n \in \mathbb{N}} |x(n)|.$$

$$c_{00} = \left\{ x \in F(\mathbb{N}, \mathbb{K}) \mid \exists k \in \mathbb{N} \text{ with } x(n) = 0 \quad \forall n \geq k \right\}$$

$$c = \left\{ x \in F(\mathbb{N}, \mathbb{K}) \mid (x(n)) \text{ is convergent} \right\}$$

$$c_0 = \left\{ x \in F(\mathbb{N}, \mathbb{K}) \mid (x(n)) \rightarrow 0 \text{ as } n \rightarrow \infty \right\}$$

$$c_{00} \subseteq c_0 \subseteq c \subseteq l^\infty$$

Let $x \in l^p \Rightarrow \sum_{i=1}^{\infty} |x(i)|^p < \infty$

$$\Rightarrow x(n) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow x \in c_0$$

Thus.

$$c_{00} \subseteq l^p \subseteq c_0 \subseteq c \subseteq l^\infty$$

Diagram illustrating the relationships between the spaces and their corresponding norms:

- c_{00} is associated with $\|\cdot\|_\infty$ and $\|\cdot\|_p$.
- l^p is associated with $\|\cdot\|_p$.
- c_0 is associated with $\|\cdot\|_\infty$.
- c is associated with $\|\cdot\|_\infty$.
- l^∞ is associated with $\|\cdot\|_\infty$.

Remark.

$$1 \leq p < r < \infty$$

$$l^p \subsetneq l^r$$

Ex.

$$1 \leq p < r < \infty$$

then, $l^p \subsetneq l^r$.

•

$$\|x\|_r \leq \|x\|_p.$$

Let $x \in l^p \Rightarrow |x(n)| < 1$ for large n .
 $|x(n)|^r < |x(n)|^p$
 $x \in l^r$.

We have show,

$$\left(\sum_{i=1}^{\infty} |x(n)|^p \right)^{1/r} \leq \left(\sum |x(n)|^p \right)^{1/p}.$$

$$a_n = |x(n)|^p \quad \alpha = \frac{r}{p} > 1.$$

Use,

$$\sum a_n^\alpha \leq \left(\sum a_n \right)^\alpha \quad (\text{complete})$$

Ex. $x \in \bigcup_{1 \leq p < \infty} l^p$

then

$$\|x\|_p \rightarrow \|x\|_\infty \text{ as } p \rightarrow \infty.$$

Try
later.

• Example of NLS.

$$\Omega \neq \emptyset.$$

$B(\Omega, \mathbb{K})$ = set of all $\mathbb{K}$ -valued
fns define on Ω that
are bdd.

if $x, y \in B(\Omega) \rightarrow$ linear space.
 $\lambda x \in B(\Omega)$
 $\Rightarrow x + y \in B(\Omega)$

$$x \in B(\Omega) \Leftrightarrow \sup_{t \in \Omega} |x(t)| < \infty.$$

Define $x \mapsto \|x\|_\infty$.
 $\|x\|_\infty = \sup_{t \in \Omega} |x(t)|.$

Note that $\|\cdot\|_\infty$ defines a norm.

then $(B(\Omega), \|\cdot\|_\infty)$ is a nls.

In particular $\Omega = [a, b]$

$$\begin{aligned} C(\Omega, \mathbb{K}) &= C(\Omega) \\ &= C[a, b] \end{aligned}$$

If $x \in C[a, b]$ then x is bounded and

$$x \in B[a, b]$$

$$\therefore C[a, b] \subseteq B([a, b])$$

We define $\|\cdot\|_\infty$ in $C[a, b]$

$$\cdot \mathbb{R}[a, b] \subseteq B([a, b])$$

$\hookrightarrow \|\cdot\|_\infty$

$$C[a, b] \begin{cases} \|\cdot\|_\infty = \sup_{t \in [a, b]} |\cdot| \\ \|\cdot\|_p \end{cases}$$

$$= \left(\int_a^b |x(t)|^p dt \right)^{1/p}.$$

$$\left(\int_a^b |x(t) + y(t)|^p dt \right)^{1/p} \leq \left(\int_a^b |x(t)|^p dt \right)^{1/p} + \left(\int_a^b |y(t)|^p dt \right)^{1/p} \quad 1 \leq p < \infty$$

Banach space.

Complete NLS = Banach space.

Ex.

1. $(\mathbb{K}, |\cdot|)$ is a Banach space.

2. $(\mathbb{K}^n, \|\cdot\|_p)$? ?

$$\|x\|_p \begin{cases} \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} \\ \max_{1 \leq i \leq n} |x(i)| \end{cases}$$

Qn. Is $(\mathbb{K}^n, \|\cdot\|_p)$ a Banach space?

- $(\mathbb{K}^n, \|\cdot\|_\infty)$ is a Banach sp.
- Equivalent criteria.

$(\mathbb{K}^n, \|\cdot\|_\infty)$ is B.S.

Let $(x_m)_{m \in \mathbb{N}}$ be C. Seq in $(\mathbb{K}^n, \|\cdot\|_\infty)$

ie $\forall \epsilon > 0 \exists N_\epsilon$ s.t

$$\|x_m - x_k\|_\infty < \epsilon, \forall m, k \geq N_\epsilon$$

ie $\max_{1 \leq i \leq n} |x_m(i) - x_k(i)| < \epsilon, \forall m, k \geq N_\epsilon$

ie $|x_m(i) - x_k(i)| < \epsilon, \forall m, k \geq N_\epsilon$
 $\forall i \in \{1, \dots, n\}$

$$\begin{aligned} x_1 &= (x_1(1), x_1(2), x_1(3), \dots) \\ x_2 &= (x_2(1), x_2(2), x_2(3), \dots) \\ x_3 &= (x_3(1), x_3(2), \dots) \end{aligned}$$

ie, $(x_m(i))_{m \in \mathbb{N}}$ is a C.S in $\mathbb{K}$ for each $i = 1, 2, \dots, n$.

Since $\mathbb{K}$ is a B.S $\exists \alpha_i \in \mathbb{K}$ s.t $x_m(i) \rightarrow \alpha_i$ as $m \rightarrow \infty$ for each $i = \{1, 2, \dots, n\}$

Let $x = (x(1) \dots x(n))$
 $= (\alpha_1 \dots \alpha_n) \in \mathbb{K}^n$.

Then $\|x_m - x\|_\infty = \max_{1 \leq i \leq n} |x_m(i) - x(i)|$
 $= \max_{1 \leq i \leq n} |x_m(i) - \alpha_i|$

$\therefore \rightarrow 0$ as $m \rightarrow \infty$.

Then $(\mathbb{K}^n, \|\cdot\|_\infty)$ is a B.Space.

• $x \in \mathbb{K}^n$ then

$$|x(i)| \leq \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} \quad \forall i \in \{1, 2, \dots, n\}$$

$$\therefore \max_{1 \leq i \leq n} |x(i)| \leq \|x\|_p$$

$$\Rightarrow \|x\|_\infty \leq \|x\|_p \rightarrow \textcircled{1}$$

$$\begin{aligned} \|x\|_p &= \left(\sum_{i=1}^n |x(i)|^p \right)^{1/p} \\ &\leq \|x\|_\infty \left(\sum_{i=1}^n 1 \right)^{1/p} = n^{1/p} \|x\|_\infty \end{aligned} \quad \text{--- } \textcircled{2}$$

To gether.

$$\underline{\|x\|_\infty} \leq \|x\|_p \leq \underline{n^{1/p}} \underline{\|x\|_\infty} \quad \textcircled{3}$$